Since the request is intercepted and interpreted at Layer 7, the possibilities of adding intelligence grow exponentially. Rules can be configured to distribute traffic based on a field in the HTTP header, the source IP address or custom cookie fields, to name just a few. There are endless possibilities to make intelligent traffic distributions. For example, if the incoming request is from a smartphone, it can be sent to servers hosting mobile applications. If the request is for a URL that hosts a simple HTML-based site, it can be routed to an economical server farm. If a login cookie is not present in the request, it can be sent to a login server, avoiding loading down other busy servers.

As the Layer 7 rules bring programmability to balancing techniques, they can further be explored for the benefit of the technology operations staff. When a roll-out of a newer version is planned in an existing database server farm, the new set of servers can be configured as a separate pool to perform migration mock-tests, and can be brought online once the tests are passed. In case the roll-out experiences problems, merely switching pools back to the original settings can achieve a roll-back with minimum downtime. As another example, many mission-critical Web farms require to maintain legacy server operating systems for stability reasons, while new applications demand the latest and greatest platforms. In such cases, separate server pools can be configured for new applications, and traffic distribution can be achieved by checking Web request URLs at Layer 7.

Load balancing at Layer 7 also helps improve the return on investment (RoI) of an IT infrastructure. Consider a Web portal which caters to a high volume of users with Web pages that are content rich, with JavaScript and images. Since the scripts and images don't change quite often, these can be treated as static content, and hosted on a separate set of servers. As a result, the Web servers running important business logic use fewer resources, which means that we can accommodate more users per server, or host more applications per server, and thus reduce the effective cost of hosting. This also proves that a carefully configured Layer 7 load balancer can achieve higher application performance throughputs on a given data-centre infrastructure footprint.

Additional features in a load balancer appliance

Besides powerful traffic distribution features, most industry-grade modern load balancers also come with features which are essential to take additional tasks from the managed nodes, or other infrastructure components. SSL negotiation is one such feature that can handle heavy volumes of SSL handshaking—which would otherwise take a performance toll on Web servers. Another great feature is cookie persistence, which helps applications stick to a particular server, in order to maintain a stateful session with it.

Many new load-balancer trends provide admin features such as traffic monitoring and TCP buffering; security features such as content filtering, and an intrusion detection firewall; and also performance-based features such as HTTP caching, HTTP compression, etc. Since a load-balancing device is a front-end component in a server farm, it comes equipped with high-speed network ports, such as Gigabit Ethernet and fibre connections.

Open source load-balancing solutions

Multiple vendors provide industry-grade enterprise load-balancing solutions, such as F5 networks (BigIP), Citrix Netscaler, Cisco, Coyote Point, etc. These devices are rich in features, provide flexible rule programmability, and exhibit high performance throughput—but they do come with a price tag and support cost. For those who are interested in FOSS (Free and Open Source Software), there are multiple distributions available on the Linux platform, which offer features from simple load balancing to full-featured appliance-grade products. Let's look at three such 'most wanted' solutions.

LVS (Linux Virtual Server) is one famous solution, which has proved to be industry-grade software, and can be used to build highly scalable and available Linux cluster servers to cater to high volumes of Web requests. It comes with ample documentation, which helps build a load-balanced farm, step by step, and can be found at http://www.linux-vs.org.

Ultra Monkey is another interesting solution, which provides fail-over features in addition to basic load balancing: if one load-balancer device fails, the other can take over to provide device-level fault tolerance. It supports multiple Linux flavours such as Fedora, Debian, etc, and can be found at http://www.ultramonkey.org.

Another powerful, but lesser-known implementation is Crossroads for Linux, which is a TCP-based load balancer providing a very basic form of traffic distribution. The beauty of this product is that its source code can be easily modified to serve just one task, such as DNS or Web balancing, without any bells and whistles—thus achieving a very high performance for that single purpose. It can be found at http://crossroads.e-tunity.com.

Configuring Layer 7 rules on a load balancer is an art, and needs a deep understanding of networking protocols and server operations. Features of load balancers can also be used as an aid to the operations and maintenance tasks. END

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